

README

Submission metadata, access instructions, resources, and contribution summary.

Project Information

Field	Details
Project	Instacart image-based reorder prediction checkpoint
Primary code repository	GitHub repo
Instacart dataset	Kaggle dataset link
Product image source	Open Food Facts
Task type	Binary classification of high-reorder vs. low-reorder products from product images

Group Members

Name	Role / Notes
Jordan Ehrman	Lead assembly of the writeup and broader project framing; repository owner reflected in the provided GitHub link.
Colton	Implemented the core notebook experiments, model training runs, and image-loading pipeline used in the checkpoint.
Amelia	Provided conceptual support and identified Open Food Facts as a practical external image source for the checkpoint.

Data Access Instructions

The Instacart tabular data used for this checkpoint comes from the public Kaggle dataset linked above. The image side of the checkpoint uses Open Food Facts as an external image source.

The current workflow first computes product-level reorder statistics from Instacart purchase history, then labels the top quartile as high-reorder and the bottom quartile as low-reorder, excluding the middle 50 percent. Product names are then used to query Open Food Facts, and candidate matches are ranked by string similarity before selecting the top acceptable image.

Because the checkpoint code uses external image URLs rather than bundling all image files directly into the repository, internet access may be required at runtime for image retrieval unless the matched image set is cached locally in advance.

Implementation Note on Augmentation

The project proposal describes light training-time augmentation such as flips, slight rotations, and mild brightness or contrast jitter. However, the uploaded training notebook currently uses resizing, tensor conversion, and normalization only. The README therefore describes augmentation as a planned extension to the baseline preprocessing pipeline rather than as something fully implemented in the present checkpoint notebook.

Resources Used

- PyTorch and torchvision for model construction and training
- Pillow for image loading and RGB conversion
- scikit-learn for dataset splitting and evaluation metrics
- Instacart Market Basket Analysis dataset on Kaggle
- Open Food Facts website / API / image infrastructure
- GitHub repository for code submission and versioning

Notes Before Submission

Code should be submitted through GitHub rather than uploaded directly to Canvas, consistent with the assignment instructions.

This version intentionally omits student ID fields, per the requested revision.